

Collaborative Editor with Distributed Networking & AI Assistance

GROUP MEMBERS

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1. Project Scope

The project aims to develop a cloud-based real-time collaborative document editor with AI-powered assistance, enabling seamless multi-user collaboration. The system will offer features similar to Google Docs, integrating AI for writing suggestions and chat-based assistance. Users can co-edit documents, communicate via an embedded chat system, and leverage AI to enhance writing efficiency. The platform will prioritize performance, security, and scalability, ensuring a smooth user experience.

2. Functional & Non-Functional Requirements

Functional Requirements (FRs):

1. User Management:

- Users can sign up, log in, and manage their profiles.
- Role-based access (e.g., admin, editor).

2. Document Management:

- Create, edit, and delete documents.
- Share documents with other users.
- Track changes in documents.

3. Collaboration & Chat:

- Multi-user real-time editing with cursor tracking.
- Live chat system for discussion.
- AI-powered message generation in chat.

4. **AI Assistance:**

- Smart suggestions for words and paragraphs.
- AI-generated content from images.
- AI-enhanced search for document content.

5. **Task & Notification System:**

- Users can create, assign, and manage tasks.
- In-app notifications for updates.

6. **Access Control:**

- Authentication and authorization for document access.

Non-Functional Requirements (NFRs):

1. **Performance:**

- The editor should handle at least **10 users** editing a document simultaneously without delays.
- Actions like saving, or sending messages should have a minimum response time of **2 seconds**.

2. **Scalability:**

- The system should be designed to support **20 users** in the future without requiring major changes.
- It should be able to handle **20 documents** efficiently.

3. **Usability:**

- The interface should be user-friendly and easy to navigate, even for beginners.
- Users should be able to understand how to use the system without extensive training.

4. **Availability & Reliability:**

- The system's availability is approximately **99.89%**, with a downtime of **10 hours per year**.

5. **Security:**

- The system should have secure authentication methods to prevent unauthorized access.

6. Maintainability & Modularity:

- The code should be organized and modular, making it easy to update or add new features.
- Developers should be able to fix bugs or improve features without affecting the entire system.